

Structural Adaptations to Climbing in the Gibbon Hand¹

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In the past, the unusual features of the gibbon hand have been viewed as structures whose function is to keep the thumb out of the way during brachiation. Recent behavioral and anatomical observations, however, indicate that climbing is as important to gibbons in their high, exclusively arboreal niche as brachiation, and that many peculiarities of the gibbon hand actually represent structural adaptations to climbing. The functional reevaluation of the gibbon hand makes it clear that complex and often contradictory selection factors are responsible for phenotypic manifestations and that the classification of species into locomotor types, such as brachiator, semibrachiator, and quadruped, misrepresents the species' total locomotor adaptation and obscures the complex of selective factors responsible for structural evolution.

THE GIBBON HAND is remarkable for its long, slender form and the deep separation of the thumb from the palm. Schultz (1936:435-436) describes these unusual proportions, and Biegert (1963:245) shows the peculiar shape of the hand in illustrations. Traditionally, these characteristics have been explained as adaptations to brachiation. The form of the hand in *Colobus* and *Ateles* has been attributed to this same selective factor, even though the thumb is reduced or lost entirely in these genera and the fingers do not show the extreme elongation found in *Hylobates* (Schultz 1930; Marzke 1971). It has been argued that the deeper separation of the gibbon thumb allows it to be folded out of the way during brachiation and thus to function like the reduction of the thumb (Straus 1942). But in terms of evolution and adaptation, this comparison is unrealistic since in *Hylobates* a whole anatomical complex—involving the form of bones, joints, and muscles—has been modified and maintained by selection, whereas in *Colobus* and *Ateles* total or near total reduction of structures has taken place.

The resolution of this problem involves fundamental issues in the interpretation of the evolutionary relationships between the structures of a particular form and their functions. These issues will be discussed

briefly before we proceed to a discussion of the gibbon hand.

The first issue concerns the complex nature of selection pressures and the concept of multiple functions. Mayr (1963:6) expressed the importance of this issue to the understanding of evolutionary biology: "Virtually every component of the phenotype is recognized as a compromise between opposing selection pressures." Tinbergen (1965:532-533), too, has implicated these factors in an attempt to explain why natural selection has not produced maximally effective social signals, i.e., structures and movements that are as effective as experimentally optimized "supernormal" stimuli. In response to this question, he states that "one answer, often proposed but rarely fully documented experimentally, is found in the fact that many different selection pressures acting on an animal are often contradictory or conflicting, and that selection (which rewards or penalizes the phenotype, not isolated characters) has produced a compromise." He adds that "it is becoming increasingly clear that these interactions of selection pressures are of immense complexity, and also that, without detailed functional study, one cannot even begin to understand adaptedness."

The theory of multiple functions and complex selection pressures is equally rele-

vant to the study of evolving locomotor structures. For example, primate hands serve multiple functions. The diversity of grips has been described by Napier (1956, 1960, 1961) and for a much wider range of primates by Bishop (1962). In chimpanzees, hands are used in brachiation, knuckle-walking, feeding, grooming, and tool-using (Goodall 1965; Tuttle 1967). Hands potentially perform various functions, and the survival value of form cannot be accurately evaluated by correlations with single, isolated functions.

A second issue of importance to the form-function relationship concerns the utility of classifying and labeling primate species into categories of locomotor types. Napier (1963) recognized the problems inherent in this approach in his attempts to deal with locomotor adaptations as diverse as those of the gibbon and spider monkey, but adding the category *semibrachiator* does not seem to have adequately expressed the tremendous differences in the total locomotor adaptations of these genera. The important point is that no one word accurately represents the total repertoire of behaviors; for an understanding of the nature of adaptations, anatomical structures must be correlated with all the movements used by individuals under natural conditions (Oxnard 1963, 1967; Oxnard and Neely 1969).

With these general principles in mind, the relationships between the functions of the gibbon hand and the form of its component structures will be reconsidered. General problems relating to the interpretation of form and function and to the labeling of locomotor types will be considered further in the discussion.

MATERIALS AND METHODS

Observations of living gibbons were made at the San Francisco Zoological Gardens, the Knowland Zoo in Oakland, and the Field Station for Behavior Research, University of California, Berkeley. Fitting a tame female *Hylobates lar* with a belt and leash made it

possible to observe her locomotion in trees outside of and adjacent to the cage area. Although the animal was somewhat restricted, the belt and leash did allow a wide range of normal locomotor activities. Series of long jumps, for instance, were not possible, but single jumps up to thirty feet were observed.

A cluster of four eucalyptus trees, ranging from about twenty to forty feet high and from five to eight inches in diameter five feet above ground level, were used for this purpose. No branches projected from the trunks of these straight, nearly vertical trees lower than twenty-five feet in the tallest and ten feet in the shortest. This aspect of the study trees is quite similar to that of many trees in the natural habitat (Ellefson, personal communication).

Additional behavioral information available in the excellent study of Carpenter (1940) and through discussions with John O. Ellefson (who from March, 1964, to December, 1965, studied the ecology and behavior of gibbons in the Malay Peninsula) has also been incorporated into this evaluation of the gibbon hand.

Anatomical data in this paper have been drawn largely from existing literature (Schultz 1930; Straus 1942; Marzke 1971; Tuttle 1967). Marzke's exhaustive work represents the most recent comprehensive review of literature concerning the anatomy of primate hands and, as such, provided most of the descriptive material.

Joint manipulations led to interesting discoveries about the potential range of movement at the junction of the trapezium and first metacarpal. Such manipulations were especially informative when combined with a close study of gibbon skeletal materials (*Hylobates lar*) from collections of the Department of Anthropology, University of California, Berkeley, and dissections of one siamang (*Symphalangus syndactylus*) and one gibbon (*Hylobates sp.*) hand.

DISCUSSION

Several primate species show reduced thumb structures. *Ateles* and *Colobus*, for

example, show both skeletal and muscular reductions of the thumb. The three great apes likewise have reduced thumb structures in the sense that the long flexor tendon to the terminal segment of the thumb is either rudimentary or absent and does not function to flex the end of the thumb (Straus 1942:236; Tuttle 1967:65). These species have been arbitrarily classified as brachiators, and on the basis of this classification a cause-effect relationship has been suggested between brachiation and reduced thumb structures, even though reductions of the thumb are also found in one highly terrestrial species, *Erythrocebus patas*. This same cause-effect relationship between brachiation and thumb reduction has been applied to the interpretation of the form of the gibbon hand. In the gibbon, a deep cleft extending to the base of the first metacarpal provides a long free portion to the thumb. Passive or negative functions have been attributed to this structural configuration, and it has been suggested that the cleft functions to keep the thumb out of the way during brachiation, either by folding it into the cleft or by folding across the proximal palm (Straus 1942:239). The accuracy of classifying together genera and species with such widely divergent locomotor adaptations as the great apes, *Ateles*, *Colobus*, and *Hylobates* is highly questionable; the extension of this correlation and hypothetical cause-effect relationship to the interpretation of the gibbon thumb is even more unacceptable.

The gibbon hand in no way compares with the kinds of reduced thumb structures characteristic of the great apes, *Ateles* and *Colobus*. The deep muscle of the forearm supplying the gibbon thumb is well developed, and a nearly independent and fully functional tendon is present. Moreover, the gibbon thumb is not reduced in size. Although LeGros Clark (1959:51) describes the gibbon thumb as short, it is short only in relation to the greatly elongated metacarpals II-V which form the palm of the hand. The thumb of a ten pound gibbon is, in fact, very nearly as long as the thumb of a 150 pound

man. It is relatively one of the longest primate thumbs (Schultz 1930:382) and is supplied by a complete set of well-developed and fully functional intrinsic and extrinsic muscles (Straus 1942:239).

To interpret the peculiar features of the gibbon thumb as structures primarily adapted to keeping the thumb out of the way is to neglect several anatomical and behavioral facts. The gibbon thumb is actively used in various postural and locomotor functions. Carpenter, for example, repeatedly observed gibbons using one hand and both feet to climb up tree trunks while holding food in the other hand. He also noticed gibbons climbing vertical trunks of large diameter and concluded that the wide gaps between thumb and fingers and between great and other toes were of the greatest advantage in such activities (Carpenter 1940:70-75). Ellefson's observations confirm those of Carpenter. He especially observed gibbons using this climbing pattern while descending small tree trunks to catch insects near ground level. At such times, the thumb is often held widely abducted from the hand during climbing even when the diameter of the trunk appears to be too large for the hand to grasp (Fig. 2); this use of the hand may provide additional friction by allowing the terminal segment of the thumb to be pressed into irregularities in the bark surface (Ellefson, personal communication).

Figures 1 to 3 illustrate some characteristic gibbon climbing activities. In Figure 1, the animal supports its weight by grasping with the thumb in opposition to the fingers. The deep cleft between the thumb and the radial side of the hand enables the animal to encircle the trunk with thumb and fingers. The free left hand reaches with the thumb poised in opposition to the fingers in preparation for securing another hand hold. When climbing trees of relatively large diameter that lack horizontal branches for hand holds, the gibbon grasps the trunk between thumb and fingers and pulls itself up from above while at the same time grasping between great and other toes and thrusting itself upward from below with the hip and



Figure 1. A typical climbing posture with the thumb and great toe grasping a trunk in opposition to the fingers and toes. Note the thumb of the left hand poised in readiness for the next hand hold.

leg (Fig. 1). At times an individual gibbon depends entirely on the grasping thumb of only one hand and pulls its entire weight up to the next hand hold without the use of the other hand or of either foot. More commonly, the adult female at the Field Station for Behavior Research used the toe and thumb in the situation shown in Figure 3, where her weight was supported by the grasping hand and foot of one side while she stretched laterally with the other hand and foot to reach small end branches of adjacent trees. This postural capacity offers great selective advantages in feeding since fruits and berries grow at the ends of small terminal branches (Avis 1962; Marzke 1971; Grand, in preparation). New leaves, for which gibbons show a marked affinity, are also found at the ends of branches (Ellefson, personal communication).

Activities like these are seldom observed in captivity because gibbon cages are designed for brachiators in the traditional sense of the term and usually provide no apparatus adequate for the type of normal climbing discussed here. Nevertheless, behavioral observations both in the natural habitat and in more diverse laboratory settings indicate that this type of active, agile climbing is a frequent and important part of gibbon locomotor and feeding adaptations and that its importance is reflected in both the external form and internal structures of the gibbon hand.

For example, close observations of climbing sequences verify the importance of the extrinsic flexor muscle of the thumb, for during the arm-pull phase of climbing the terminal segment of the thumb, which is supplied solely by the extrinsic flexor, has been observed to flex on the proximal phalanx just as the animal lifts its weight from a previous support (Fig 4).

The remarkable mobility of the thumb also indicates that this structure is neither reduced nor passive. Mobility is achieved in part by the cleft between thumb and second metacarpal and is further enhanced by the modified ball and socket joint between the first metacarpal and trapezium, a feature unique to gibbons (Marzke 1971).

Moreover, observations and joint manipulations show that the thumb does not, in fact, flex across the proximal palmar surface as suggested by Straus (1942:239). When the animal swings from small branches, the only folding that takes place in the thumb is at the metacarpophalangeal and interphalangeal joints, movements that do not utilize the cleft. During brachiation on branches of large diameter, the thumb is either held against the side of the hand as Straus (1942:239) suggested or actually folded onto the dorsum of the hand (Ellefson, personal communication).

A closer look at the anatomy shows that the thumb cannot be folded across the proximal palm because this movement is prevented by the structure of the articulative surfaces and the ligaments between the



Figure 2. Use of the thumb in grasping a vertical trunk of large diameter.

thumb and wrist. The configuration of the trapezium, the base of the first metacarpal, and the joint ligaments are such that in a

position of full abduction—with the metacarpal of the thumb approaching an angle of ninety degrees with the plane of the remain-

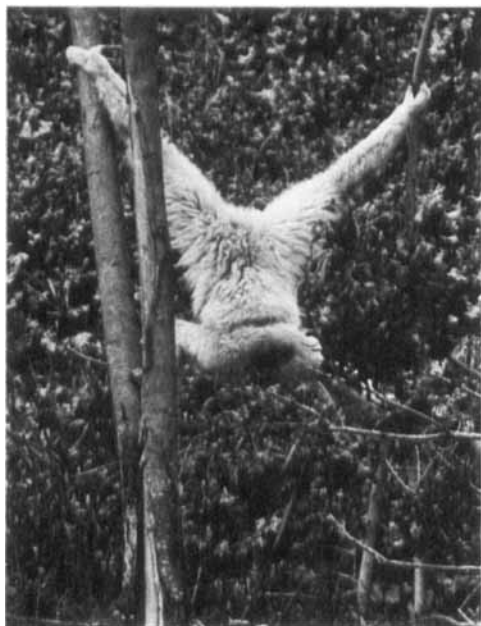


Figure 3. A common feeding posture among gibbons. The left hand and foot support most of the animal's weight while the other hand and foot are used to gather leaves and fruit from small terminal branches.



Figure 4. During climbing, the long flexor muscle to the terminal segment of the thumb supplies an important component of the total grasping power.

ing metacarpals—flexion, extension, and rotation of the first metacarpal on the trapezium are inhibited. This condition is made possible by the arrangement of ligaments around this joint. A loose fibrous capsule encloses the carpometacarpal joint between the thumb and wrist. Two ligaments lie within this capsule at the distal edge of the joint. The dorsal ligament of this pair is shorter than the palmar. As abduction proceeds, the dorsal ligament tenses before the palmar and causes the first metacarpal to rotate medially in opposition to the lateral rotation caused by the trapezium (Fig. 6). At ninety degrees abduction, both ligaments become fully tensed and prevent flexion, extension, and rotation (Figs. 5, 6). The final result of this complex movement is that the thumb now lies widely abducted from the index finger and second metacarpal but with its palmar surface almost completely in

opposition to the palmar surface of the index finger. Additional rotation at the metacarpophalangeal joint completes this opposition; this final degree of rotation probably takes place passively during grasping. At full abduction, the thumb is stabilized at the trapeziometacarpal joint, and flexion and extension are greatly reduced. In this position, the extrinsic flexor to the terminal segment of the thumb in conjunction with the flexors of the fingers supplies an important grasping force.

In positions of slight or intermediate abduction, the trapeziometacarpal articulation enables the animal to rotate the first metacarpal so that when he grasps a substrate of small diameter the palmar surface of the thumb can be placed against the radial side of the hand in the region of the metacarpophalangeal joint of the index finger. The gibbon commonly uses this grasp when moving about or feeding among small end branches.



Figure 5. The bones, ligaments (L), and joint capsule (C) of the gibbon thumb (dorsal view) in full adduction.

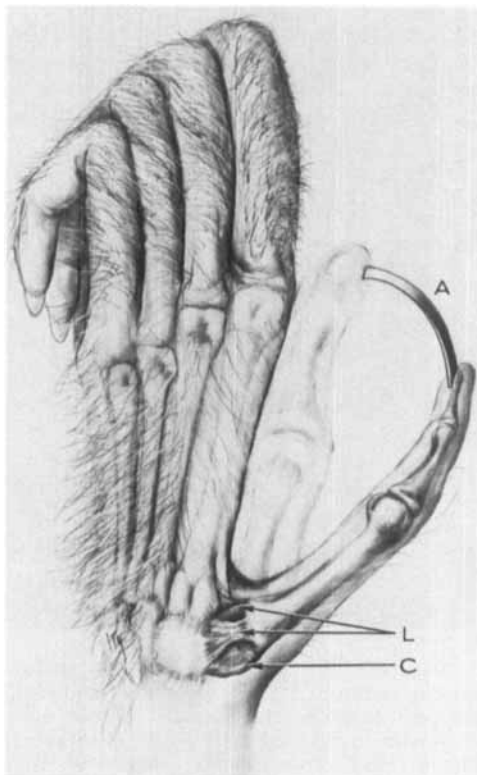


Figure 6. Dorsal view of the gibbon hand in full abduction, the ligaments (L) within the joint capsule (C) tense and prevent rotation, flexion, and extension. As the thumb is abducted, it rotates laterally on the trapezium as the thumb moves through the arc A. At the end of abduction, the lateral ligament tenses sooner than the medial and produces a few degrees of medial rotation.

The structure of the gibbon thumb allows powerful grasping actions during a wide variety of postural and locomotor activities. The survival value of these behavioral capacities can be closely correlated with feeding habits as mentioned above (Avis 1962; Marzke 1971). But the relevance of Schultz's (1956) data to this problem should not be underestimated. His reports of high frequencies of healed bone fractures among wild apes suggest the ever-present danger of falling and the potential importance of falls

as an effective selection agent among populations of strictly arboreal animals living high in the canopy. In a series of 260 wild adult gibbons, thirty-three percent had healed bone fractures. The almost automatic habit among gibbons of grasping a nearby object when at rest, even though securely seated, represents a behavioral and anatomical adaptation to this danger.

CONCLUSIONS

Our understanding of the fossil record and of living primates can be greatly ex-

panded by the discovery of basic correlations between form, function, and adaptation. Accurate observations and descriptions of the anatomy and habits of living species in their natural environments constitute the only evidence on which these correlations can be based.

The emphasis in this paper has been on the form of the hand and its use in climbing activities. But the form of the gibbon hand has evolved in response to complex functions and cannot be explained by selection for a single behavioral capacity. Although behavioral and anatomical evidence indicates a high degree of correlation between the thumb and climbing functions, neither the form of the thumb in its entirety nor the total form of the hand can be explained in terms of this single function. The anatomy of the gibbon hand and of the brain that controls the hand allows a variety of behaviors, and selection may operate differentially on the specific components of this behavioral complex through time.

Viewed in this way, explaining anatomical structures in evolutionary terms involves the evaluation of the adaptive relevance or survival value (Tinbergen 1965) of specific behaviors and the structures that perform these behaviors (Prost 1965:1212-1213; Ripley 1967). Among hylobatids, climbing and brachiating represent behaviors of enormous adaptive importance. Schaller (1963:475) notes the extensive geographical distribution of gibbons. The successful occupation of a high, exclusively arboreal habitat (Ellefson, personal communication) depends on the evolution of anatomical structures that can effectively utilize this zone. The important functions of feeding and grooming have been included in the evolution of locomotor specializations that enable the gibbon to grasp with speed and skill. In fact, selection pressures for manipulation may counteract those structural variations within populations that are mechanically more effective in climbing. To evaluate the relative contributions of specific behavioral capacities and of the selective factors acting on these capacities to the form

of the gibbon hand represents an extremely complex and challenging problem in the interpretation of the adaptive function of behaviors and their structural bases. A comprehensive analysis of these relationships must await ecological studies that will describe the normal ranges and frequencies of locomotor and social activities in a diversity of species and habitats (Ripley 1967). Prost (1965:1213) and Ripley (1967) have also emphasized the need to distinguish between frequency and adaptive significance, and the evaluation of this distinction depends on the accuracy of ecological information.

NOTE

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